

Education

Arizona State University

Bachelor of Science in Mechanical Engineering | GPA: 3.50

Tempe, AZ

Expected May 2028

- Dean's List (Fall 2024, Spring 2025, Fall 2025)

- Relevant Coursework: Mechanics of Materials, Structure and Properties of Materials, Statics, Thermodynamics, Dynamics, Circuits

Marcos de Niza High School

Diploma | GPA: 3.54

Tempe, AZ

May 2024

- Extracurriculars: Chamber Orchestra (8 years), Varsity Team Captain for Football, Baseball, and Track & Field

Job Experience

Institute for Digital Inclusion Acceleration

Digital Navigator

Chandler, AZ

January 2025 – Present

- Instruct over 300 weekly visitors in hardware and software fundamentals, facilitating thousands of hands-on technology interactions across diverse age groups.
- Develop and deliver structured curriculum for nine distinct software platforms, including Swift, TinkerCAD, and SpheroEDU, standardizing the instructional process for the center.
- Operate and maintain Flashforge 3D printers using Orca Slicer, troubleshooting hardware failures and optimizing print parameters to ensure consistent output quality.
- Mentor community members on digital literacy and device safety, translating complex technical concepts for users with varying levels of technical proficiency.

Wingstop

Cook

Tempe, AZ

June 2025 – August 2025

- Cooked food
- Bagged food

Sonic Drive-In

Shift Lead

Tempe, AZ

April 2022 – August 2024

- Managed the efficiency of workers and food preparation for shifts.
- Delivered accurate orders, achieving a 103% customer satisfaction rating.
- Ensured food safety and quality standards in a high-volume environment.

Tilly's

Sales Associate

Tempe, AZ

June 2024 – August 2024

- Employee of the Month (July 2024) for exceeding sales targets.
- Increased sales with effective visual merchandising and personalized service.
- Trained new associates, enhancing team collaboration and performance

AZCEND

Volunteer

Chandler, AZ

September 2023 – February 2026

- Helped with data management, inputting data to help measure program success.
- Aided in the distribution of resources to communities through events like Operation Santa.
- Organized the layout of branches with both the management of others and physical labor.

Project Experience

AI-Driven Quantitative Equity Research & Portfolio Optimization Terminal

May 2026 – Present

- Architected a multi-page Streamlit application utilizing the OpenBB SDK to aggregate institutional-grade market data, SEC EDGAR filings, and real-time news APIs into a centralized, low-latency research dashboard.
- Engineered a custom 5-factor quantitative scoring model (Value, Momentum, Quality, Size, Volatility) that calculates advanced risk-adjusted metrics—including Sharpe, Sortino, and Calmar ratios—to systematically evaluate equity upside and downside risk.
- Integrated a local Large Language Model (Ollama/Llama 3.1) to synthesize unstructured news sentiment and extract legal risk factors from SEC 10-Q filings, generating a composite AI confidence score for targeted assets.

- Designed a modular backend pipeline to integrate the Sophtron API for real-time brokerage holdings and PyPortfolioOpt for Modern Portfolio Theory (MPT) rebalancing, transforming raw alpha signals into actionable, capital-constrained trade recommendations.

VR Rapid Prototyping Application

Tempe, AZ

Prototyping Lead | Modeling Team

March 2025 – December 2025

- Developed a virtual reality prototyping environment for Meta Quest headsets using the Unity engine, integrating over 100 interactive 3D assets.
- Directed a team of modelers in Unity, establishing technical workflows and resolving engine blockers to deliver the functional prototype by the end of the semester.
- Partnered with faculty to design the underlying curriculum framework, establishing the technical foundation for future integration into introductory engineering courses.

Documentation Lead

January 2025 – March 2025

- Authored technical reports, Gantt charts, and meeting minutes to track design milestones and maintain project version control.
- Standardized reporting templates to improve progress tracking and delivered a final technical package detailing project scope, design rationale, and future development recommendations.

Mental Health Robotics System for Isolated Environments

Tempe, AZ

Design and Prototyping Lead

August 2024 – December 2024

- Programmed an Arduino Uno microcontroller to process inputs from ultrasonic distance and photoresistive light sensors, enabling semi-autonomous environmental navigation.
- Fabricated the physical robot chassis using laser-cut wooden components, ensuring precise mechanical fit and structural rigidity for the electronic modules.
- Executed successful demonstration trials of the semi-autonomous prototype, validating sensor accuracy and mechanical stability under operational conditions.

Shock Absorber

August 2025 – December 2025

- Designed and assembled a shock absorber for a car using SolidWorks
- Created technical part drawings

Computer-Aided Design Modeling Projects

August 2024 – December 2025

- Developed SolidWorks and Fusion 360 part files and produced fully dimensioned engineering drawing sheets that follow standard drafting conventions, reinforcing skills in technical communication and manufacturability.
- Applied CAD best practices and design intent principles to strengthen 3D visualization abilities and prepare for advanced coursework in mechanical design and manufacturing.

Skills

Technical: SolidWorks, Fusion 360, MATLAB, TinkerCAD, Arduino Circuitry & Coding, Unity Game Engine, 3D Printing

Professional: Leadership, Communication, Adaptability, Team Collaboration, Problem Solving, Critical Thinking

Microsoft Suite, and Google Workspace

Other: Sewing, Roofing, Construction, Childcare, Automotive Repair, Mathematics Tutoring, Music Theory, Guitar, Violin, Piano

SolidWorks

- 6 months of experience, almost have CSWA, and working towards CSWP

Fusion 360

- 6 months of experience

MATLAB

- 1 year of experience

Unity Game Engine

- 7 months of experience

Sewing

- 2 years of experience

Piano

- 4 years of experience

Guitar

- 3 years of experience

Violin

- 8 years of experience

Activities and Campus Involvement

Society of Hispanic Professional Engineers	Tempe, AZ
Outreach Director of STEM Events	<i>March 2026 - Present</i>
Boxing at ASU	Tempe, AZ
General Member	<i>January 2026 - Present</i>
Salsa at ASU	Tempe, AZ
General Member	<i>March 2026 - Present</i>

Certifications/Awards

Dean’s List	Fall 2024, Spring 2025, Fall 2025
•Maintained above a 3.5 GPA while taking 12+ credit hours	